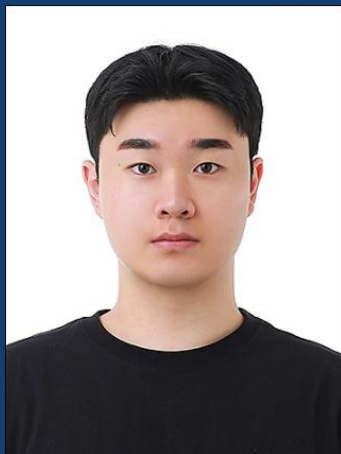




Gwanwoo Ju

Jeonju, Republic of Korea

gwanwoo7835@jbnu.ac.kr
gwanwoo7835@gmail.com

+82 10-6275-7835

Language

Korean, English

Education

03/2019 – 02/2025

Bachelor's degree, Department of Mineral Resources and Energy
Engineering, Jeonbuk National University, South Korea

03/2025 – Current

Integrated Ph. D Course, Department of Climate, Environment and Energy,
Jeonbuk National University, South Korea

Research project

04/2024 – 09/2024

- Canada-South Korea Climate Action Partnership: Emission Reduction
Paths and Effects Evaluation

08/2024 – 07/2026

- Development of Screening Criteria for Enhanced CO₂ Injection and
Optimal Injection Well Patterns into the aquifer

03/2025 – 12/2026

- Global Talent Development Program for Innovative Smart Carbon Capture
and Storage Operations through Artificial Intelligence and Digital Twin
Integration

Working Experiment

08/2025– 03/2026

- Visiting Student Researcher (VSR) in Department of Geomatics
Engineering, University of Calgary

Training

- Underground Carbon Storage Modeling Using STOMP CO₂ Simulator
(KIGAM, 2023)
- Underground Hydrogen Storage Simulation & CO₂ EOR Modeling
(CMG, 2024)
- Eclipse Black Oil Reservoir Simulation (SLB, 2024)
- Underground CO₂ Storage Modelling & Geomechanics Modelling
(CMG, 2024)
- Eclipse Compositional Simulation (SLB, 2025)
- Introduction to CO₂ Flow Assurance in transportation and injection
(SLB, 2025)
- Reservoir Simulation using Petrel RE and INTERSECT (SLB, 2026)

Publications

- Lee, Y., **Ju, G.**, Kim, M., Kim, J., Lee, Y.*. Physics-Informed Machine Learning for Rapid Assessment of CO2 Injectivity and Plume Migration: From Surrogate Modeling to Field-Scale Application (*Under review*)
- Lee, Y., Dehghan, M. J., Kim, J. W., Ali, M. Y., **Ju, G.**, Oh, J. Wavenumber Correlation of Gravity and Magnetic Anomalies for Constraining Salt-Basement Structure in Offshore Abu Dhabi (*Under review*)

Patent

- **Ju, G.**, Lee, Y. Apparatus and Method for Evaluating CO2 Geological Storage Feasibility in Aquifers (*Patent Pending*)